Stroke Risk Prediction Model

A clinical referral decision through statistical inference



Course: Modern Applied Statistics for Data Science

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1. Introduction

Stroke is one of the leading causes of death and long-term disability worldwide, yet much of its burden is preventable through early identification and management of modifiable risk factors such as hypertension, elevated blood glucose, cardiac disease, and smoking (GBD 2019 Stroke Collaborators, 2021). Primary-care clinicians cannot refer every patient for prevention counseling, because capacity is finite and each referral uses a slot another patient could have had. The task is to prioritize patients under uncertainty. When the event rate is around 5%, a classifier can look highly accurate and still be clinically useless, so we treat the problem as one of statistical inference and decision-making.

2. Research Question and Decision

Primary question. Which patient characteristics are associated with stroke, by how much (with quantified uncertainty), and at what predicted-probability threshold does referral do more good than harm when the base rate is about 5%?

The decision-maker is a primary-care physician deciding whether to refer a patient for stroke-prevention counseling. A false negative leaves a high-risk patient unreferred and risks a preventable stroke. A false positive uses up a scarce counseling slot. The two errors have different costs and prevalence is low, so Bayes' rule implies that even a screen with 95% sensitivity and 95% specificity has a low positive predictive value: at this dataset's 4.87% base rate it is about 49%, meaning roughly half of all referrals would be false alarms. The referral threshold is therefore the quantity of interest, and raw accuracy is not.

3. Data and Exploratory Analysis

We use the real, publicly available, de-identified Stroke Prediction Dataset (fedesoriano, 2021), which has 5,110 patient records and 12 columns. We do not use the synthetic Kaggle Playground Series S3E2 data as the primary source, because a distribution cannot be justified from the data-generating process when the data were synthetically resampled.

Column	Description (observed range or counts)
id	Record identifier; dropped before analysis
age	Age of the patient in years (0.08 to 82)
hypertension	Whether the patient has hypertension (0/1)
heart_disease	Whether the patient has heart disease (0/1)
avg_glucose_level	Average blood glucose (55 to 272); units are not stated, but the values match mg/dL
bmi	Body-mass index (10.3 to 97.6); 201 values missing

smoking_status	never smoked 1,892 / formerly smoked 885 / smokes 789 / Unknown 1,544
gender, work_type, Residence_type, ever_married	Demographic and context variables (gender includes one "Other" record)
stroke	Outcome label, whether the patient had a stroke (0/1); 249 strokes

Initial EDA. Strokes are rare: 249 of 5,110 patients (4.87%), which is the base rate used throughout. Two columns have missing information. BMI is missing for 201 patients (3.9%), and smoking status is "Unknown" for 1,544 (30.2%), which the dataset description defines as information that was unavailable. The missingness is not random (Figure 1b). Patients with missing BMI had a stroke rate of 19.9%, against 4.3% for patients with BMI recorded, so dropping them would remove 40 of the 249 strokes (16%) and leave a lower-risk sample. We will compare a complete-case analysis with multiple imputation and report both. We keep "Unknown" smoking status as its own category, which retains 30% of the records.

The data also include 856 patients under 18 (16.8%), who account for only 2 of the 249 strokes; we will check whether results change when children are excluded and report both versions. A single record has gender "Other", which cannot support a group estimate, so we exclude it from gender comparisons only. Average glucose has two peaks (Figure 1a): most patients sit around 80 to 90 mg/dL, with a second, smaller group near 200 to 230 mg/dL, a range consistent with diabetes. BMI has 13 values above 60, with a maximum of 97.6. We will examine and report these outliers and will not delete them to force a distribution.

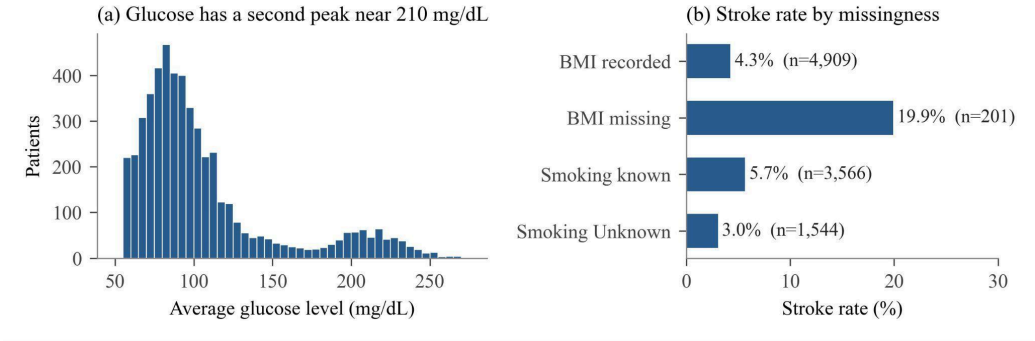


Figure 1. (a) Distribution of average glucose level for all 5,110 patients. (b) Stroke rate among patients with and without missing BMI and smoking information; the overall rate is 4.87%.

4. Distributional Modeling

For each continuous variable we first argue why a distribution is plausible from how the data are generated, and then test the fit. Average glucose is strictly positive and right-skewed, so a Gamma or log-normal model is a more natural candidate than a Normal one. Figure 1a also shows a second peak, so a single Gamma or log-normal curve may fit poorly; if it does, we fit a two-component mixture and explain what the misfit means for any analysis that relies on it. We assess fit with quantile-quantile (Q-Q) plots and a formal goodness-of-fit test (Kolmogorov-Smirnov or Anderson-Darling), and we report heavy tails as a finding.

5. Estimation and Uncertainty

Every estimate comes with a confidence interval and an effect size; we never report a bare point estimate or a lone p-value. For group differences we report standardized effect sizes such as Cohen's d or the difference in proportions, and for the model we report odds ratios with intervals. Where normality is doubtful, as with skewed variables or small subgroups (there are only 249 strokes), we use percentile bootstrap intervals with about 10,000 resamples (Efron & Tibshirani, 1993). To check that the intervals can be trusted, we simulate repeated samples from the fitted distributions and count how often the intervals cover the known value. We interpret a 95% interval through the procedure's long-run coverage: across many repeated samples, about 95% of intervals built this way would contain the true value.

6. Hypothesis Testing

We state each null (H_0) and alternative (H_1) hypothesis here, before running any test. For each one we choose a test, state and check its assumptions, and fall back to a rank-based alternative when they fail. We report every test we run, significant or not, to avoid selective reporting and p-hacking (Wasserstein & Lazar, 2016), and we pair each p-value with an effect size and its confidence interval so that significance is not confused with practical importance.

Hypothesis	Test (with assumption check / fallback)
H_0: mean glucose is equal in stroke and non-stroke patients; H_1: it differs	Welch's t-test; Mann-Whitney U if normality fails
H_0: hypertension and stroke are independent; H_1: they are associated	Two-proportion z-test or χ^2 test of association
H_0: age has no effect on stroke risk ($\beta = 0$); H_1: risk increases with age ($\beta > 0$)	Sign and size of the age coefficient, plus a trend test

7. Statistical Modeling and Validation

The primary model is logistic regression, chosen because its coefficients become odds ratios that we can report with confidence intervals and tie back to the hypotheses (Hosmer, Lemeshow, & Sturdivant, 2013). We compare it with a simple baseline, such as an intercept-only or single-predictor model. We validate out of sample with a stratified held-out set and k-fold cross-validation that keeps the 4.87% class ratio in every fold, and we assess calibration with a reliability diagram and a calibration test (Collins et al., 2015). We then turn the model into the referral decision. Sweeping the referral threshold, we show how PPV, sensitivity, and expected clinical cost change under the base rate, and we recommend the threshold that minimizes expected cost given the unequal error costs. Accuracy and AUC are reported for completeness only. So that the results lead to action, we group risk factors by whether a patient can realistically change them:

Category	Features
Modifiable	Hypertension, average glucose level, BMI, smoking status
Non-modifiable	Age, gender, prior heart disease

8. Ethics and Data Handling

The Kaggle page lists the licence as "Data files © Original Authors" and describes the source as confidential, for educational use only (fedesoriano, 2021). We use the data for this course only and credit the author as the page requests. Because the original source is not disclosed, we cannot say which population the patients represent, and we do not generalize our estimates to any specific clinic or country.

The records are already de-identified, and we make no attempt at re-identification. We drop the id column before analysis, since it is the only per-record identifier and has no analytic use. We examine whether model errors fall unevenly across age and gender groups and report any disparities alongside the aggregate performance. Wherever we impute, transform, or exclude data (the missing BMI values, the single "Other" gender record, extreme BMI values, or patients under 18), we report the result before and after the change.

9. Documentation and Reproducibility

All results come from a scripted, version-controlled pipeline that goes from the raw CSV to every figure and reported number with one command. We fix random seeds for every step that uses randomness, record them in the repository, and pin every package to a specific version. A short data card records the source, licence, cleaning steps, and known limitations.

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